

1.1.1 Web Development

Front-end developers

Front-end developers focus on what users interact with on the website. They use HTML, CSS, and JavaScript to bring web designs to life. Key tools and technologies include:

- CSS preprocessors like LESS or Sass for efficient style management.
- JavaScript frameworks such as AngularJS, React, Ember.js, and Vue.js have revolutionized how developers build dynamic and responsive web applications.
- Libraries like jQuery.
- Git and GitHub for version control.

They also use on-site search engine optimization (SEO) to make sure the website ranks high in search engine results.

Back-end developers

Back-end developers handle the data and server-side functions of web applications. Essential tools and skills include:

- Programming languages like PHP, Python, Java, and C#.
- Frameworks such as Ruby on Rails, Symfony, and .NET that streamline building robust server-side applications.
- Database management systems, including MySQL, MongoDB, and PostgreSQL.
- Security and authentication technologies such as OAuth and Passport for protecting data integrity and user privacy.

Front-end developer	Back-end developer
It is Part of website User Can see Code and Interact it.	It is brain of website that Code never visible to End User
Run on the client side browser	Run on the server side browser
HTML, CSS, JavaScript (and frameworks like Bootstrap, React).	PHP, Python, Java, Node.js and Databases like MySQL, MongoDB.
Designs the layout, look and feel of the website.	Handles the logic, database, and server operations.
Limited Security	Advanced Security

1.2 Client–Server Architecture

1.2.1 Definition & Concept

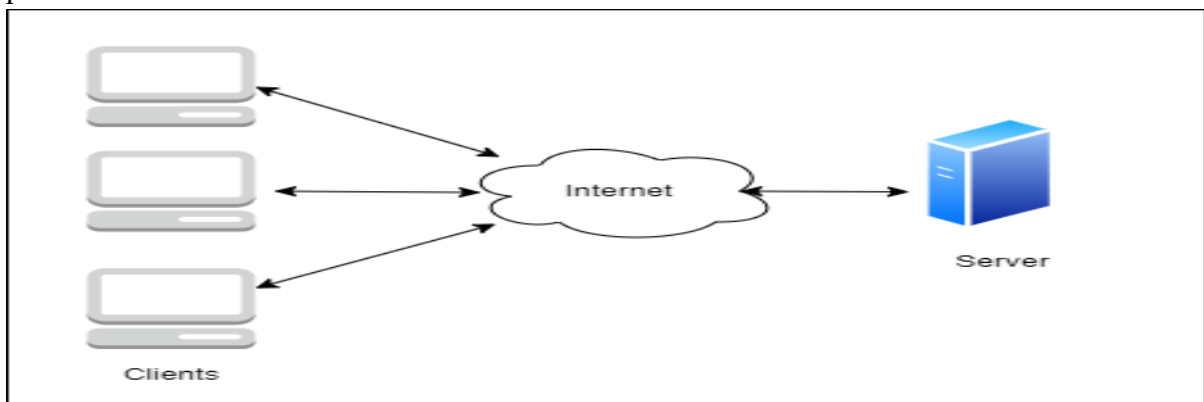
Client-Server Architecture is a model where tasks are divided between two entities:

(i) Client = Request

(ii) Server = Response

A computing model where the client requests services and the server processes and responds.

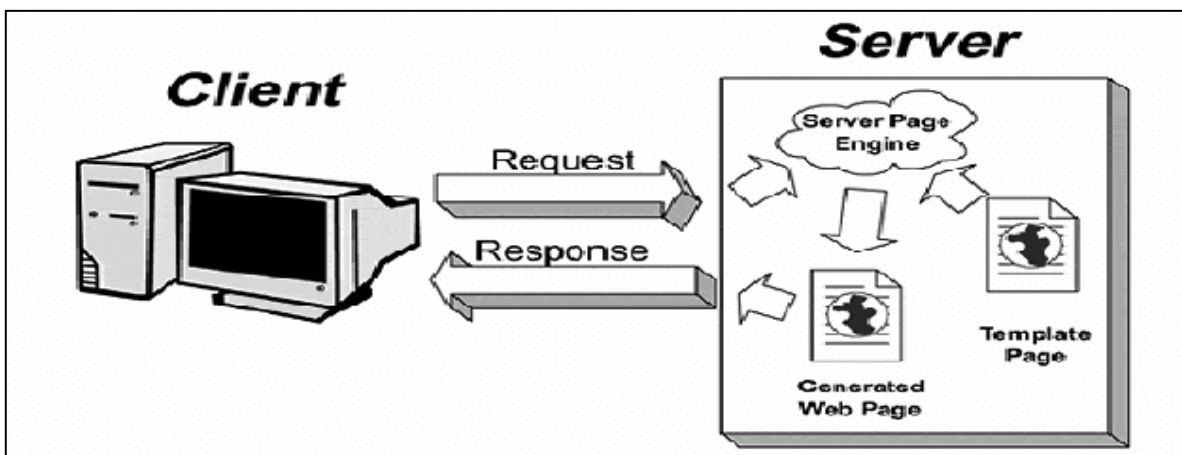
Client-Server Model is a distributed architecture where clients request services and servers provide them



a) **Client:** An application that requests services from the server, such as data retrieval, storage, calculations, or other functions.

b) **Server:** An application that processes client requests, sends responses, or performs specified actions. The server and client may reside on the same machine or different devices across the network. Effective Communication occurs via predefined protocols like HTTP, FTP, or SMTP.

1.2.2 How client requests and server responds



1.2.3 Types of Client-Server Architecture

1. 2-tier Architecture

Client directly communicates with Server.

Two tier architecture primarily has two parts, a client tier and a server tier. The client tier sends a request to the server tier and the server tier responds with the desired information.

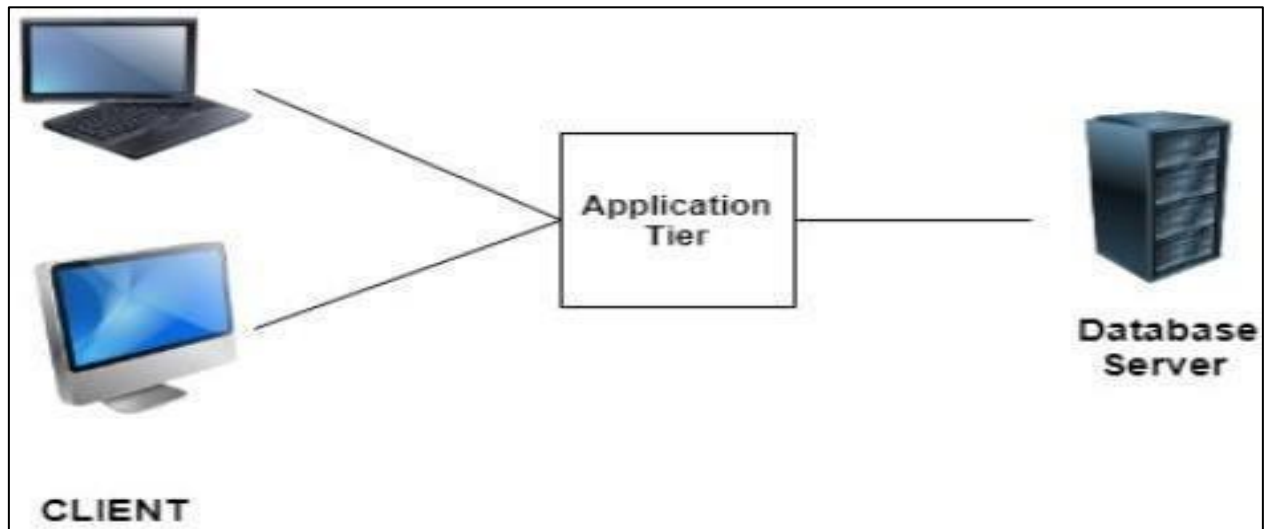


2.3-tier Architecture

Three tier architecture has three layers namely client, application and data layer.

Client → Application Server → Database.

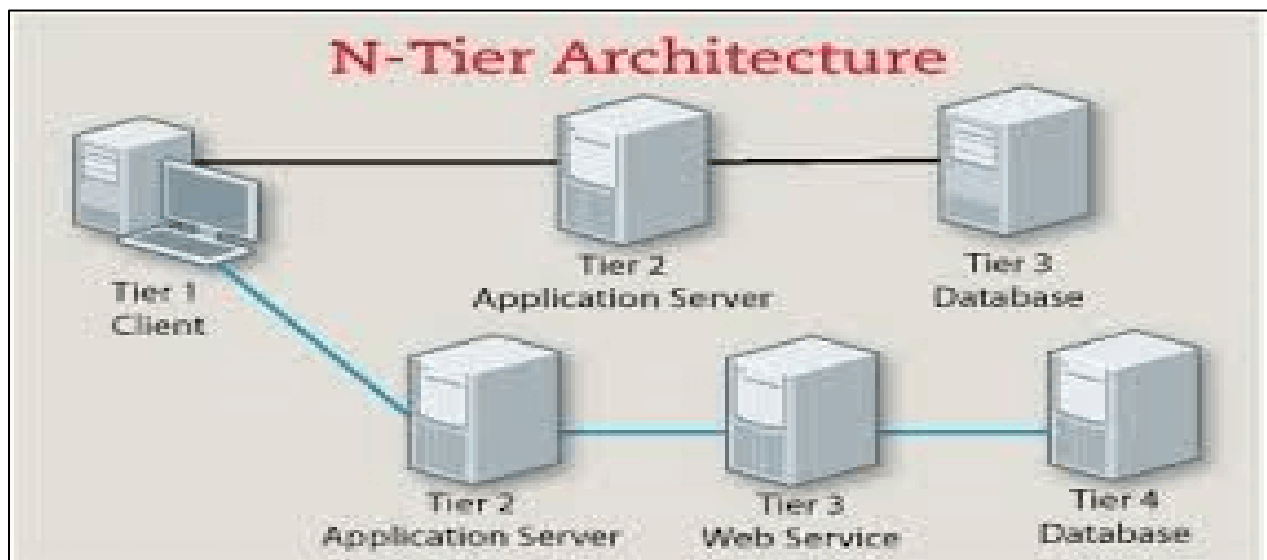
client layer is the one that requests the information. In this case it could be the GUI, web interface etc. The application layer acts as an interface between the client and data layer. It helps in communication and also provides security. The data layer is the one that actually contains the required data.



3. N-tier Architecture

N-tier architecture is also called Distributed Architecture or Multi-tier Architecture. Multiple layers of servers for scalability.

It is similar to three-tier architecture but the number of the application server is increased and represented in individual tiers



1.4. Introduction to Text Editors

A Text Editor is software used to write and edit code.

Text editor is a program that lets the user edit text. Whenever you're able to type some text in the input, you are editing the text, and that input is somehow a text editor. By this definition, we can call the HTML input element a basic text editor.

1.4.1 Basic text editors

The simplest form of text editors is the basic one. In a basic text editor, you can create and edit plain text without any styling or formatting.

- TextEdit (Mac)
- Notepad (Windows)
- Nano (Linux)
- HTML textarea (Web)

1.4.2 Integrated Development Environments

IDEs are some sort of special text editors that are designed for the programming and software development purposes. Syntax highlighting, intelligence and code completion, version controlling and debugging are the benefits of integrated development environments.

- Visual Studio
- Eclipse
- Jet Brains Products like PyCharm and WebStorm

1.4.3 Difference between Text Editor & IDE

Text Editor: Lightweight, mainly for writing code (Notepad, Sublime, Atom).

- **IDE:** Advanced, includes debugging, compilation, version control (VS Code, IntelliJ).

Feature	Text Editor	IDE (Integrated Development Environment)
Definition	A simple tool to write and edit plain text or code.	A complete software suite that provides tools for writing, testing, and debugging code.
Main Purpose	Only for writing and editing text/code.	For software development (coding, compiling, debugging, project management).
Complexity	Lightweight and simple.	Heavy, feature-rich, and advanced.
Examples	Notepad, Notepad++, Sublime Text, VS Code (when used just for editing).	Eclipse, PyCharm, IntelliJ IDEA, Visual Studio, Android Studio.
Features	Syntax highlighting, basic editing.	Code editor + Debugger + Compiler/Interpreter + Build automation + Version control integration.
Usage	Best for small coding tasks or quick edits.	Best for large projects and full application development.
Resource Requirement	Low system resources (RAM, CPU).	Requires more system resources.